Dual ODE: Spatial–Spectral Neural Ordinary Differential Equations for Hyperspectral Image Super-Resolution

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***Abstract*— Significant advancements have been made in hyperspectral image (HSI) super-resolution with the development of deep-learning techniques. However, the current application of deep neural network architectures to HSI super-resolution heavily relies on empirical design strategies, which can potentially impede the improvement of image reconstruction performance and introduce distortions in the results. To address this, we propose an innovative HSI super-resolution network called dual ordinary differential equations (Dual ODEs). Drawing inspiration from ordinary differential equations (ODEs), our approach offers reliable guidelines for the design of HSI super-resolution networks. The Dual ODE model leverages a spatial ODE block to extract spatial information and a spectral ODE block to capture internal spectral features. This is accomplished by redefining the conventional residual module using the multiple ODE functions method. To evaluate the performance of our model, we conducted extensive experiments on four benchmark HSI datasets. The results conclusively demonstrate the superiority of our Dual ODE approach over state-of-the-art models. Moreover, our approach incorporates a small number of parameters while maintaining an interpretable model design, thereby reducing model complexity.**

***Index Terms*— Dual ordinary differential equations (Dual ODEs), hyperspectral image (HSI) super-resolution, ordinary differential equations (ODEs).**

I. Introduction

HYPERSPECTRAL images (HSIs) encompass both spectral and spatial information, which can be widely utilized in diverse fields such as mineral exploration [1], medical diagnosis [2], plant detection [3], target detection [4], anomaly detection [5], change detection [6], terrain classification [7], tracking [8], disaster monitoring [9], urban planning [10], and defense applications [11].

However, the obtained HSI always has high spectral resolution but low spatial resolution due to hardware limitations. Since the HSI super-resolution method provides ways to obtain HSIs with high-resolution (HR) both in the spectral and spatial domains, it becomes the research focus within the remote-sensing community. One of the HSI super-resolution methods is to reconstruct an HSI with high spatial resolution directly from its counterpart with low spatial resolution, which is also the research focus of this study.

Different kinds of HSI super-resolution methods have been proposed, among which deep-learning-based methods become the dominant method (the details can be seen from the related works) [12], [13], [14], [15]. Within the deep-learning-based method, residual neural networks with shortcuts [16] have demonstrated their effectiveness for HSI super-resolution since they can address the issue of gradient vanishing problems caused by increasing network depth. However, two limitations impede the further improvement of the existing residual neural network based on the HSI super-resolution method. On the one hand, previous models based on residual modules, such as ResNet, rely solely on artificially designed patterns, which are limited in terms of their form and flexibility. On the other hand, as the network depth increases, achieving convergence requires more advanced training techniques. Directly applying traditional residual modules to predict pixel-level fine-grained information in HSI super-resolution may not yield optimal results. Therefore, there is a need for innovative approaches that provide reliable theoretical guidance and explainability, allowing for more efficient and interpretable designs for HSI super-resolution networks.

To tackle the challenges mentioned above, we propose a new approach named dual ordinary differential equation (Dual ODE) that utilizes ordinary differential equation (ODE) methods to design a spectral–spatial feature extraction network for HSI super-resolution. The Dual ODE approach consists of two crucial components: a spatial ODE block for extracting spatial information and a spectral ODE block for capturing internal spectral features. By incorporating these ODE blocks, we enhance the effectiveness of HSI super-resolution by

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harnessing the strengths of ODEs in capturing spatial–spectral characteristics while utilizing fewer parameters. Additionally, by leveraging different ODE solvers, we can explore the potential of higher-order numerical methods to improve the feature extraction capability of the network. This approach combines the advantages of ODE methods to design a spectral–spatial feature extraction network. The aim is to enhance performance while minimizing the computational burden and cost of the model.

Moreover, we seamlessly integrate these designed Dual ODE blocks into the spectral and spatial feature extraction stages of the HSI super-resolution task. This integration facilitates the extraction of both spatial and spectral features, resulting in improved performance in HSI super-resolution. To validate our models, we conduct thorough evaluations on benchmark HSI datasets. The results demonstrate that our methods surpass state-of-the-art approaches while maintaining comparable parameters.

In summary, this article presents three significant contributions.

1) We propose a new HSI super-resolution approach called Dual ODE that utilizes ODE techniques to design a spectral–spatial feature extraction network. By leveraging the power of ODEs and optimizing parameter usage, our method excels in capturing both spatial and spectral characteristics. This leads to a significant enhancement of HSI super-resolution performance.
2) We introduce two key components, namely the spatial ODE block and the spectral ODE block, which play crucial roles in extracting spatial information and capturing internal spectral features, respectively. The integration of these ODE blocks enables our approach to effectively extract both spatial and spectral features, leading to superior performance in HSI super-resolution. Moreover, our Dual ODE blocks seamlessly integrate into the spectral and spatial feature extraction stages of the HSI super-resolution task, providing a comprehensive and efficient solution.
3) Through extensive evaluations of benchmark datasets, we demonstrate the superiority of our approach over state-of-the-art methods. Our Dual ODE method achieves remarkable improvements in HSI super-resolution performance while maintaining comparable parameters and computational costs. These results solidify the significance of our contributions and highlight the potential of our approach for advancing HSI super-resolution research.

II. Related Works

A. Traditional Methods for HSI Super-Resolution

Traditional methods for HSI super-resolution always treat the HSI super-resolution task as an optimization problem, which incorporates different image priors as constraints first and then finds the optimum solution by solving the optimization problem. The obtained optimum solution is the expected HR HSI. For this purpose, different kinds of image priors are constructed including total-variant prior, sparse prior, and low-rank prior. For example, Liu et al. [17] addressed the challenge of limited training samples by designing a trainable grouped joint tensor dictionary that accurately maps HR HSIs to low-resolution (LR) HSIs. Gelvez et al. [18] presented a non-data-driven deep image prior-based HSI super-resolution method using interpretable deep blocks and a novel loss-function strategy, demonstrating superior performance compared to state-of-the-art approaches.

However, the above methods first transform an HSI from a tensor into a matrix, which does not fully utilize the spatial-spectrum characteristic within HSI data. For this purpose, some methods that can preserve both spatial and spectrum structures are proposed, among which the tensor-based method is representative since it inherently can reflect the structure of a high-order tensor. For example, Yokoya et al. [19] presented the coupled nonnegative matrix factorization hybrid method for fusing HSI data. In a different direction, Zhang et al. [20] proposed an unsupervised adaptive learning framework, adapting general image priors to specific HSIs. Building upon this, He et al. [21] introduced the coupled tensor ring decomposition method, achieving improved super-resolution. Additionally, Akhtar et al. [22] proposed a sparse representation-based HSI super-resolution method, interpreting high-spatial-resolution images using extracted spectra.

Despite their promising performance, the reliance on specific image priors and accurate degradation assumptions limits the generalization ability of these methods.

B. Deep-Learning-Based Methods for HSI Super-Resolution

In recent years, deep-learning methods have achieved remarkable success in natural image super-resolution, primarily due to the strong representation capabilities of convolutional neural networks (CNNs) [23], [24], [25], [26], [27]. Inspired by the success of CNNs in high-level computer vision tasks like image classification, Dong et al. [28] introduced a CNN-based algorithm for image super-resolution, establishing CNNs as the mainstream approach in the field [29], [30]. These CNN-based methods aim to learn a mapping function between the LR and HR images through supervised learning. Prominent examples include SRCNN [28], EDSR [30], and SRGAN [31], among others. Inspired by these methods, different HSI super-resolution methods are proposed. For example, Li et al. [12] proposed an HSI super-resolution method based on depth-wise convolution. Li et al. [14] introduced a novel groupwise deep recursive residual network to enhance super-resolution performance. Yuan et al. [32] proposed an HSI super-resolution approach based on transfer learning, leveraging knowledge from natural images to improve HSI super-resolution. Jia et al. [13] proposed a spectral–spatial network for HSI super-resolution, which effectively enhances spatial resolution while preserving spectral information. Ran et al. [15] further proposed a general CNN fusion framework with multiscale guidance for HR images via recursive and progressive strategies, which improves the quality of image reconstruction while reducing the parameters of the model.

Despite the advantages of the existing deep-learning-based methods for HSI super-resolution, the network structure of

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these methods is designed based on empirical experience, which relies on empirical design strategies and lacks interpretability. For this reason, the existing super-resolution method can only lead to suboptimal performance and may impede the applications in reality. How to construct deep neural network structures that can be interpreted by mathematics or physics is still an unresolved problem.

C. Methods Based on Differential Equation Dynamics

ODE has proven to be an effective way to theoretically design the structure of deep neural network structure, which has good interpretability as well as memory-efficient. Weinan [33] first revealed the intriguing connection between neural network architecture and numerical differential equations. They demonstrated that high-dimensional nonlinear functions can be modeled using continuous dynamic systems, interpreting residual networks as a form of differential equations.

Researchers have then expanded on this insight by investigating the connection between traditional networks [16], [34], [35], [36] and various numerical discretizations of differential equations [37]. Chen et al. [38] introduced neural ODEs (NODEs) for efficient image classification and generation, leading to a new era of ODE-based methods that have revolutionized various visual tasks across different domains [39], [40], [41], [42], [43]. Paoletti et al. [41] extended ResNet to Neural ODE for HSI classification, capturing temporal dynamics. He et al. [40] applied an ODE-inspired design to single image super-resolution (OISR), revealing ResNet's potential in enhancing super-resolution and providing a deeper understanding of its behavior. Furthermore, recent studies have demonstrated the effectiveness of incorporating multiple ODE functions to enhance the performance of NODEs. Researchers such as Chalvidal et al. [44], Massaroli et al. [45], and Yoon et al. [46] have explored innovative approaches involving time-dependent variables, enabling the combination of multiple ODEs and leading to further improvements in model capabilities and performance. In theory, when the time step Δt described ODE function is sufficiently small, a residual unit can be effectively modeled as an ODE. It is indeed that Δt cannot be chosen arbitrarily small in practical applications, which will influence the connection between ResNets and the forward Euler discretization of an ODE to a certain extent [47], [48]. Nevertheless, the above research has demonstrated that the ODE-based method has advantages in lots of applications such as classification and time-series analysis even when utilizing a nonarbitrarily small Δt. The introduction of NODEs and the subsequent exploration of their applications have reshaped the landscape of deep learning, highlighting the potential of ODE-based methods in addressing complex visual tasks and providing novel perspectives for network design and performance enhancement.

Since the ODE-based method is a broadly new topic for image super-resolution tasks, there are few works to date. Although ODEs can be directly applied to HSI super-resolution, introducing ODEs for HSI super-resolution has two limitations.

1) The representation ability of the original ODE is limited, since it cannot learn all possible homeomorphisms of input data space, and therefore prone to quickly saturating in terms of performance even as the number of layers increases. As a result, this drawback cannot guarantee that the original ODE obtains good performance for HSI data which has complex structures.
2) Though few works attempt to introduce ODEs for image super-resolution tasks, there is no other work for HSI super-resolution based on ODEs. One of the reasons behind this is the lack of suitable ways to represent spatial–spectral structures with ODEs. To address these two problems simultaneously, we propose the Dual ODE method for HSI super-resolution. First, we introduce the recently proposed polymorphic NODE as a new solver instead of the original NODE method, which can effectively improve the representation ability of ODEs by using a time-evolving mixture weight on multiple ODE functions. More importantly, we specially constructed polymorphic Dual NODE based on polymorphic NODE, which takes full account of spectral and spatial characteristics within HSIs. As a result, the proposed method is expected to obtain better performance compared with other ODE-based methods.

III. Proposed Method

In this section, we present the framework of the proposed Dual ODE for HSI super-resolution, inspired by recent advancements in deep learning and the principles of dynamical systems. Initially, we present the constructed network structure for HSI super-resolution in Section III-A and then introduce the Dual-ODE block with time-evolving mixture weight in Section III-B. Following this, in Section III-C, we provide the theoretical midpoint Euler method as the solution of the Dual-ODE block. Ultimately, we present the Runge–Kutta family of ODEs for HSI super-resolution in Section III-D.

A. Dual ODE Proposed for HSI Super-Resolution

The schematic can be seen from Fig. 1. The spectrum is first divided into three overlapping parts and then each part serves as the input for each branch. In addition, each branch is composed of three modules including the HSFE head, HSFE body, and HSFE tail.

1) HSFE Head: The HSFE head is responsible for the initial shallow feature extraction from the input image. Each branch has a specific convolutional layer separately.

2) HSFE Body: The HSFE body plays a crucial role in the overall architecture of the proposed Dual ODE for HSI super-resolution. It consists of a series of spatial–spectral feature extraction modules, as illustrated in Fig. 1 (body). The spatial–spectral feature extraction modules aim to extract deep spatial–spectral features from the input HSI. As depth increases, these features become increasingly distinctive and informative, allowing for a more precise reconstruction of the HR HSI. The spatial ODE solver and the spectral ODE solver within the HSFE body are utilized for spatial and spectral feature extraction, respectively. For the spatial ODE solver,

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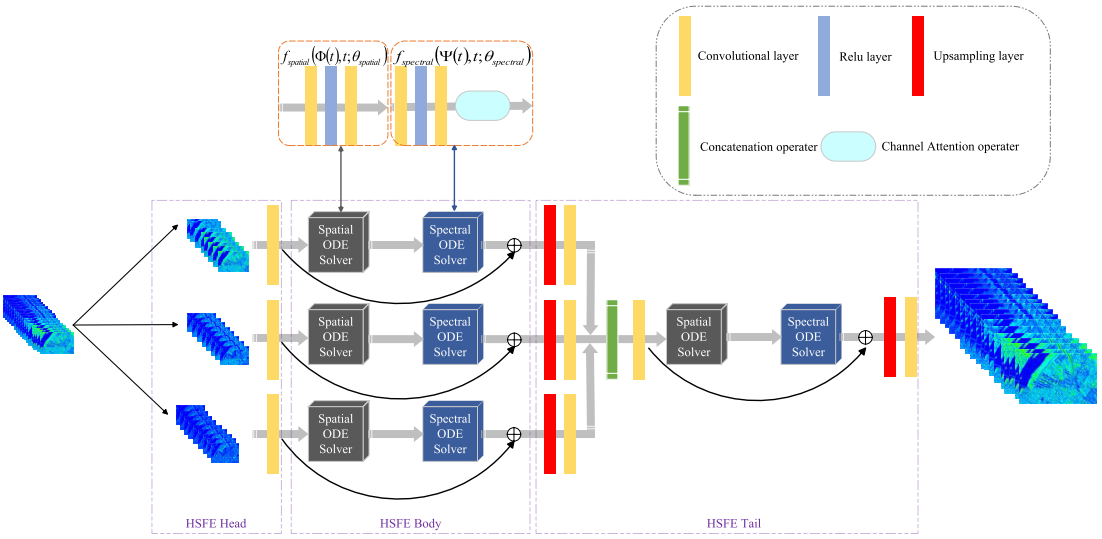


Fig. 1. Overall architecture of the proposed Dual ODE. The network architecture has three branches, and each branch comprises three modules including the HSI super-resolution feature extraction (HSFE) head, HSFE body, and HSFE tail. In addition, the spatial ODE and the spectral ODE solvers are embedded in the HSFE body and HSFE tail to extract deep features.

TABLE I

SPECIFIC PARAMETER SETTINGS FOR EACH OF THE CONVOLUTIONAL LAYERS. FOR THE CAVE AND HARVARD DATASETS, $n_{\text{BANDS}} = 31$; FOR THE PAVIA DATASET, n_{BANDS}S = 102; FOR THE CHIKUSEI DATASET, $n_{\text{BANDS}} = 128$

Module	Sub-module	Kernel	Activation	Stride	Padding	Upsampling	Concatenation	Attention
HSFE Head	a convolutional layer	256 x 3 x 3 x n_{bands}	-	1	Yes	-	-	-
HSFE Body	spectral ODE solver	256 x 3 x 3 x 256	Relu	1	Yes	-	-	-
		256 x 3 x 3 x 256	-	-	-	-	-	-
	spatial ODE solver	256 x 3 x 3 x 256	Relu	1	Yes	-	-	-
		256 x 3 x 3 x 256	-	1	Yes	-	-	Yes(reduction=16)
	Skip connection	256 x 1 x 1 x 256	-	-	-	-	-	-
HSFE Tail	-	256 x 3 x 3 x 256	-	1	Yes	Yes	Yes (n_{bands})	-
	a convolutional layer	n_{bands} x 3 x 3 x 256	-	1	Yes	-	-	
	spectral ODE solver	256 x 3 x 3 x 256	Relu	1	Yes	-	-	-
		256 x 3 x 3 x 256	-	-	-	-	-	-
	spatial ODE solver	256 x 3 x 3 x 256	Relu	1	Yes	-	-	-
		256 x 3 x 3 x 256	-	1	Yes	-	-	Yes(reduction=16)
	Skip connection	256 x 1 x 1 x 256	-	-	-	-	-	-
	a convolutional layer	256 x 3 x 3 x n_{bands}	-	1	Yes	Yes	-	-

a convolutional layer is applied to capture spatial information, followed by a nonlinearity layer and another convolutional layer to refine the extracted features. For the spectral ODE solver, an attention layer is added at the end of the block to further enhance the representation of spectral features.

3) HSFE Tail: The HSFE tail is responsible for feature enhancement and reconstruction in the proposed Dual ODE architecture for HSI super-resolution. It receives the deep features from the HSFE body and the shallow features from the HSFE head and then combine these features and generate an HR output. The overall architecture of the HSFE tail can be seen in Fig. 1 (tail). The features obtained from the three branches of the HSFE head, HSFE body, and HSFE tail are concatenated and fed into the enhancement module of the global network. This enhancement module follows the same steps as the earlier components, ensuring consistency and continuity in the feature extraction and reconstruction process. To facilitate training and enhance the performance of the network, a long skip connection is added between the upsampling and reconstruction layer at the tail. This skip connection allows for the direct flow of information from the upsampling layer to the reconstruction layer, aiding in the preservation of important spatial and spectral details throughout the super-resolution process. Additionally, the skip connection contributes to mitigating training challenges associated with optimizing deep networks.

Through the HSFE tail, the Dual ODE architecture effectively combines and leverages the features that have been extracted from both the HSFE head and HSFE body. This integration enables the network to capture both high-frequency and low-frequency information, leading to improved reconstruction quality in HSI super-resolution tasks.

Finally, the topological details and parameter settings of the proposed Dual ODE network are summarized in Table I. The HSI input has dimensions $\mathbb{R}^{h\times w\times n_{\text{bands}}}$, where $h \times w$ represents the spatial dimensions of the input image with $h \times w$ pixel values and n_{bands} is the number of spectral bands.

B. Dual-ODE Block With Time-Evolving Mixture Weight

To provide a comprehensive understanding of the proposed Dual ODE architecture with a time-evolving mixture weight, it is necessary to introduce the background on NODEs.

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NODE models, introduced by Chen et al. [38], are latent variable models defined by the equation $(dy(t)/dt) = f(t, y(t), \theta)$, where $y(t)$ represents the vector-valued states of the forward process at time t, and $f(., \theta)$ with network parameters θ denotes the dynamics function that captures the hidden states in the neural network. In most cases, it is challenging to express $y(t)$ analytically, and numerical methods are employed. A common numerical method for solving parameterized ODEs is given by the forward-pass equation

$$y(t + \Delta t) = f_{\text{NODE}-t}(y(t); t_0) : \quad \frac{dy(t)}{dt} = f(t, y(t), \theta),\ y(0) = y_t,\ y_{t+\Delta t} = y_t \tag{1}$$

where Δt represents the discrete time step and t_0 denotes the initial state of time t. This forward pass equation is typically solved using differentiable equation solvers for ODEs with initial value problems. Common choices include the Runge–Kutta family [49] and the Euler method [50].

In contrast, the ResNet architecture, introduced by He et al. [16], employs parameterized network blocks defined by the equation

$$\frac{y_{t+1} - y_t}{\Delta t} = f_t(y_t),\ y(0) = y_t, \quad y_{t+1} = y_t, \quad \Delta t = 1. \tag{2}$$

The remarkable similarity between the ODE block described in (1) and the ResNet block described in (2) has motivated the replacement of ResNet blocks with ODE blocks. In the case of the forward Euler discretization solver, the evolution between t and $t + \Delta t$ in (1) corresponds to the evolution at layer t in ResNet. This observation enables us to interpret the discrete mapping functions f_t in (2) as the evaluations of continuous mapping functions $f(t, y(t), \theta)$ at different layers, establishing a relationship between times and layers within the neural network.

While NODE introduces a novel interpretation of a deep residual network as a continuous structure based on layer depth, it struggles with representational limitations. Specifically, it is unable to learn all possible homeomorphisms of the input data space, resulting in rapid performance saturation, even with an increasing number of layers. Expanding on this understanding, we introduce a time-evolving mixture weight that evolves alongside a separate neural ODE. This approach allows us to incorporate multiple ODE functions and adaptively combine their contributions during the learning process. As depicted in Fig. 1, the features extracted by the HSFE head are processed by the Dual-ODE block, where each $f(t)$ in the ODE numerical method is parameterized by a neural network. The Dual-ODE block computes the solution from a given initial time t_i to an intermediate time t_{i+1}. The stacking of these blocks can be seen as computing the interval $[0, T]$ from an initial time t_0 to a final time t_T. To enhance the effectiveness of network architecture design, we propose a more efficient approach for training neural ODEs. Inspired by the work of Yoon et al. [46], we introduce a time-evolving mixture weight that evolves alongside a separate neural ODE. This approach allows us to incorporate multiple ODE functions and adaptively combine their contributions during the learning process.

In particular, when using the Euler method (2), polymorphic NODEs allow for the ensemble of multiple NODEs to derive y_{t_k} from y_{t_0}, resulting in a k-layer residual network. The formulation of multiple NODEs with the Euler method is as follows:

$$\alpha_{t_{i+1}} = \alpha_{t_i} + g(\alpha_t, t_i)\Delta t,\ \ \omega_{t_{i+1}} = \text{softmax}\left(\alpha_{t_{i+1}}\right), \tag{3a}$$

$$y^k_{t_{i+1}} = y^k_{t_i} + f_k\left(y^k_{t_i}, t_i\right)\Delta t,\ \ y_{t_{i+1}} = \sum_{k=1}^{N} \omega^k_{t_{i+1}} y^k_{t_{i+1}} \tag{3b}$$

where α_t is an N-dimensional time-varying weight modeled by the dynamical system $g(\alpha_t, t_i)$, and $f_k(y^k_{t_i}, t_i)$, $1 \le k \le N$, are N dynamical systems with the same number of dimensions. Higher-order methods, such as the midpoint Euler method or the more popular Runge–Kutta method, can be used instead of the Euler method for improved performance. From a dynamical system perspective, multiple NODEs decompose a dynamical system into the product of an N-dimensional dynamical system and N independent dynamical systems. Furthermore, a detailed empirical analysis examining the sensitivity of other hyperparameters in the dynamical system has been conducted and is available in Section V.

By leveraging the Dual-ODE block and the time-evolving mixture weight, our method offers a powerful and efficient framework for training neural ODEs. It enables us to effectively capture the temporal dynamics and optimize the network architecture to achieve superior performance in HSI super-resolution tasks.

The loss function utilized in our approach draws inspiration from the research conducted by Jiang et al. [51], where they proposed a loss function specifically tailored for HSI super-resolution tasks. We adopt a similar loss function in our method to capitalize on the advantages of their approach. This loss function combines the unique characteristics of HSIs with the widely used L1 loss function for super-resolution. The specific formulas for the loss function are denoted as follows:

$$L_{\text{SSTV}}(\theta) = \frac{1}{N}\sum_{n=1}^{N}\left(\|\nabla_h I^n_{\text{SR}}\|_1 + \|\nabla_\omega I^n_{\text{SR}}\|_1 + \|\nabla_c I^n_{\text{SR}}\|_1\right), \tag{4}$$

$$L_{\text{total}}(\theta) = L_1 + \alpha L_{\text{SSTV}}. \tag{5}$$

The advantage of this loss function is that it combines both spatial and spectral correlation.

Furthermore, as discussed in Section III, deep residual networks can be interpreted as continuous dynamic systems, where the layer depth serves as the hidden states. This insightful connection between CNNs and ODEs has inspired us to develop residual modules utilizing high-order ODE numerical methods, such as Runge–Kutta methods with varying orders. The formulation of these residual modules is presented below.

C. Multiple NODEs With Midpoint Euler Method

The explicit midpoint Euler method, also known as the modified Euler method, is a numerical approximation technique that mitigates estimation errors resulting from large Euler steps. In our study, we investigate the application of this method in combination with multiple NODEs to enhance the performance of the network. Moreover, we incorporate the

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well-established Runge–Kutta method from numerical analysis to further explore the design possibilities. The resulting formulas, denoted as the midpoint Euler method (RK2) in mathematics, are presented in the following equations:

$$\alpha_{t_{i+1}} = \alpha_{t_i} + \frac{1}{2}\left(G_1^\alpha + G_2^\alpha\right),\ \ \omega_{t_{i+1}} = \text{softmax}\left(\alpha_{t_{i+1}}\right), \tag{6a}$$

$$G_1^\alpha = g\left(t_i, \alpha_{t_i}\right)\Delta t, \tag{6b}$$

$$G_2^\alpha = g\left(t_i + \Delta t, \alpha_{t_i} + G_1^\alpha\right)\Delta t, \tag{6c}$$

$$y_{t_{i+1}}^k = y_{t_i}^k + \frac{1}{2}\left(G_1^k + G_2^k\right),\ \ y_{t_{i+1}} = \sum_{k=1}^{N} \omega_{t_{i+1}}^k y_{t_{i+1}}^k, \tag{6d}$$

$$G_1^k = f_k\left(t_i, y_{t_i}^k\right)\Delta t, \tag{6e}$$

$$G_2^k = f_k\left(t_i + \Delta t, y_{t_i}^k + G_1^k\right)\Delta t. \tag{6f}$$

The RK2 block, guided by this theoretical framework, deviates from the traditional ResNet as it incorporates multiple branches, which is a commonly adopted design in popular network architectures. By harnessing the strengths of the explicit midpoint Euler method and the Runge–Kutta method, our objective is to enhance the network's capability to capture complex temporal dynamics and attain superior performance in HSI super-resolution tasks.

D. Multiple NODEs With Third-Order Runge–Kutta Method (RK3)

Inspired by the advantages offered by the midpoint Euler method and the potential for reduced local truncation errors in higher-order numerical methods for ODEs, we delve into the utilization of an RK3 combined with multiple NODEs in our experiments. The equations presented in (7) depict the graphical representation of the RK3 method.

By adopting the RK3 method, we aim to enhance the accuracy and stability of our network architecture. The incorporation of multiple NODEs further empowers the model to capture complex temporal dynamics and improve its overall performance. Through rigorous experimentation and analysis, we can gain valuable insights into the effectiveness of the RK3 method and its impact on HSI super-resolution tasks

$$\alpha_{t_{i+1}} = \alpha_{t_i} + \frac{1}{6}\left(G_1^\alpha + 4G_2^\alpha + G_3^\alpha\right),\ \ \omega_{t_{i+1}} = \text{softmax}\left(\alpha_{t_{i+1}}\right), \tag{7a}$$

$$G_1^\alpha = g\left(t_i, \alpha_{t_i}\right)\Delta t, \tag{7b}$$

$$G_2^\alpha = g\left(t_i + \frac{1}{2}\Delta t, \alpha_{t_i} + \frac{1}{2}G_1^\alpha\right)\Delta t, \tag{7c}$$

$$G_3^\alpha = g\left(t_i + \Delta t, \alpha_{t_i} - G_1^\alpha + 2G_2^\alpha\right)\Delta t, \tag{7d}$$

$$y_{t_{i+1}}^k = y_{t_i}^k + \frac{1}{6}\left(G_1^k + 4G_2^k + G_3^k\right),\ \ y_{t_{i+1}} = \sum_{k=1}^{N} \omega_{t_{i+1}}^k y_{t_{i+1}}^k, \tag{7e}$$

$$G_1^k = f_k\left(t_i, y_{t_i}^k\right)\Delta t, \tag{7f}$$

$$G_2^k = f_k\left(t_i + \frac{1}{2}\Delta t, y_{t_i}^k + \frac{1}{2}G_1^k\right)\Delta t, \tag{7g}$$

$$G_3^k = f_k\left(t_i + \Delta t, y_{t_i}^k - G_1^k + 2G_2^k\right)\Delta t. \tag{7h}$$

Furthermore, the explicit Runge–Kutta family of methods provides a flexible approach for designing network structures of any order, employing the formula (8) to update both α and y. However, it is crucial to recognize that higher-order methods often result in the generation of more complex modules. While these higher-order modules have the potential to capture finer details and intricate dynamics, it is not guaranteed that they will always lead to improved network performance

$$y_{n+1} = y_n + \sum_{i=1}^{n} \gamma_i G_i, \tag{8a}$$

$$G_1 = \text{hf}(x_n, y_n), \tag{8b}$$

$$G_i = \text{hf}\left(x_n + \alpha_i h, y_n + \sum_{=1}^{i-1} \beta_{ij} k_j\right). \tag{8c}$$

The decision to utilize higher-order methods in the design of network structures should be made after a thoughtful evaluation of the particular task and the tradeoff between model complexity and performance. In certain scenarios, a simpler first- or second-order method, such as the Euler or midpoint Euler method, may yield satisfactory outcomes. Therefore, it is imperative to conduct a comprehensive assessment and comparison of various numerical methods, taking into account their computational costs and convergence properties within the context of the specific problem domain. This evaluation will help determine the most suitable approach for the given task, ensuring an optimal balance between accuracy and efficiency.

IV. Experiments

In this section, we present our experimental findings that serve to validate the performance of the proposed method. We commence by describing the public dataset utilized in our experiments, which is elaborated on in Section IV-A. Subsequently, we introduce the evaluation metrics employed and provide a comprehensive overview of the state-of-the-art comparison methods in Section IV-B. Moving forward, in Section IV-C, we provide detailed insights into the implementation specifics of the proposed method. Finally, in Section IV-D, we present both quantitative and qualitative experimental results of various methods, aiming to showcase the superiority of the proposed method for HSI super-resolution tasks.

A. HSI Dataset

1) CAVE Dataset: The CAVE dataset comprises 32 distinct images, which can be categorized into five parts. Each HSI within the dataset consists of 31 spectral bands spanning from 400 to 700 nm, with a 10-nm interval. The spatial resolution of each HSI is 512 × 512 pixels.

2) Harvard Dataset: The Harvard dataset is divided into two parts. The first part consists of 50 real-world HSIs captured under natural daylight scenes, while the second part includes 27 HSIs captured under artificial or mixed light scenes. For this particular experiment, we selected 50 images from the first part of the dataset. Each scene in this subset contains 31 spectral bands ranging from 420 to 720 nm, with a 10-nm interval. The spatial resolution of each HSI in the dataset is 1040 × 1392 pixels.

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3) Pavia Centre Dataset: The Pavia Centre dataset was captured using the catoptrics system imaging spectrometer sensor and covers the central area of Pavia in northern Italy. Each image in the dataset has a spatial resolution of 1096 × 1096 pixels and consists of 102 spectral bands. However, 13 of these bands are noninformational due to water vapor absorption and noise, and they are discarded.

4) Chikusei Dataset: The Chikusei dataset represents an urban area in Tsukichi City, Ibaraki Prefecture, Japan, and it was captured on July 29, 2014, using the Headwall Hyperspec-VNIR-C imaging sensor. This dataset consists of 128 spectral bands, covering a spectral range from 363 to 1018 nm. The spatial resolution of the dataset is 2517 × 2335 pixels.

B. Competitors and Evaluation Matrix

In our study, we conducted a thorough comparison between the proposed method and seven other models known for their strong performance. The models included VDSR [29], EDSR [30], RCAN [52], SAN [53], 3DCNN [54], GDRRN [14], DCNN [55], OISR [40] (e.g., LF, RK2, and RK3), and SSPSR [51].

To assess the performance of the proposed method and the comparison models, we employed six widely used evaluation metrics: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), spectral angle similarity (SAM), cross correlation (CC), root mean squared error (RMSE), and erreur relative globale adimensionnelle de synthese (ERGAS). By considering these comprehensive evaluation metrics, we obtained a holistic understanding of the performance and effectiveness of the different methods in HSI super-resolution tasks.

C. Experimental Setup

1) CAVE Dataset Experiment: For this dataset, we selected 18 images as the training dataset, two images as the validation dataset, and the remaining 12 images as the test dataset (Following [51]). To improve the training stability, we cropped patches from each image within the training dataset and the validation dataset separately. Specifically, when the upsampling factor was 4, we extracted patches of size 64 × 64 with an overlap of 32 pixels from the HR image. When the upsampling factor was 8, we cropped patches of size 128 × 128 with an overlap of 64 pixels from the HR image. Then, all patches from the training dataset are utilized for training and all patches from the validation dataset are exploited for validation in the experiment. In addition, the LR HSI was generated from the HR image through bicubic interpolation.

2) Harvard Dataset Experiment: For this dataset, we followed the settings in [56] and selected 30 images, three images, and the remaining 20 images as the training dataset, validation dataset, and test dataset, respectively. Then similar to the CAVE dataset, we cropped patches from the training dataset and the validation dataset separately, from which we obtained the final training dataset and validation dataset utilized in the experiment. When the upsampling factor was 4, we cropped patches of size 128 × 128 with an overlap of 32 pixels from the HR image. When the upsampling factor was 8, we cropped patches of size 256 × 256 with an overlap of 64 pixels from the HR image. The utilized LR HSI was also generated from the HR image through bicubic interpolation.

3) Pavia Centre Dataset Experiments: For this dataset, we first extracted a subimage whose spatial size is 1096 × 715 and centered at the original Pavia Centre image. Then, for any scaling factors (i.e., upscale factor equals 4 or 8), we cropped patches of size 256 × 256 with an overlap of 32 pixels from this subimage, from which we can obtain 20 patches in total. Finally, we followed the settings in [51], and chose the first ten patches as the training dataset and the remaining ten patches as the test dataset, to guarantee that the training dataset is independent of the test dataset. In addition, the utilized LR HSI was generated from those patches through bicubic interpolation.

4) Chikusei Dataset Experiments: For this dataset, we first extracted a subimage whose spatial size is 2304 × 2048 and centered at the original Chikusei dataset. Then, we followed the settings in [51] and divided this subimage into the training dataset, validation dataset, and test dataset, respectively. Specifically, we chose the top region whose spatial size was 512 × 2048 within the subimage as the test dataset. Then, we cropped patches from the remaining region whose spatial size was 1792 × 2048 to generate the training dataset and validation dataset. When the upsampling factor was 4, we cropped patches of size 64 × 64 with an overlap of 32 pixels from the HR image. When the upsampling factor was 8, we extracted patches of size 128 × 128 with an overlap of 64 pixels from the HR image. After cropping, we chose 90% of the generated patches as the training dataset and the remaining patches as the validation dataset, to guarantee that the training dataset is independent of the validation dataset. The utilized LR HSI was also generated from those patches through bicubic interpolation.

All experiments use the ADAM optimizer [57] with an initial learning rate of 1e-4 and L1 loss function. The learning rate is decayed by 0.1 every 30 iterations, and the batch size of the model is 8.

D. Experimental Analysis

1) Results on the CAVE Dataset: In the CAVE dataset experiment, our Dual ODE method was compared with several competing approaches at 4 and 8 upscaling factors. Table II demonstrates that our method achieved the best performance in both cases. At the 8 scale, our method improved the PSNR by approximately 0.41 dB compared to the EDSR model with nearly half the parameters. Additionally, our method achieved a 0.26-dB PSNR improvement compared to the previous state-of-the-art method SAN with fewer parameters. Visually, the HSI images generated by our method exhibited fewer reconstruction errors and more details compared to the competing methods. Overall, the proposed Dual ODE method strikes a balance between model complexity and super-resolution performance, making it a superior choice for HSI super-resolution tasks.

2) Results on the Harvard Dataset: For the Harvard dataset, the proposed method demonstrates state-of-the-art

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TABLE II

QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH OTHER METHODS ON THE CAVE DATASET USING CC, SAM, RMSE, ERGAS, PSNR (DB), AND SSIM METRICS. RED INDICATES THE BEST PERFORMANCE

dataset	Model	scale	params	CC ↑	SAM ↓	RMSE ↓	ERGAS ↓	PSNR ↑	SSIM ↑
CAVE	Bicubic	4	-	0.9868	4.1759	0.0212	5.2719	34.7214	0.9277
	3DCNN [54]		0.11M	0.9876	5.0770	0.0226	5.1537	34.0291	0.9267
	EDSR [30]		24.34M	0.9933	3.4427	0.0158	3.4423	38.0107	0.9505
	GDRRN [14]		0.8M	0.9922	4.3234	0.0176	3.9044	36.5105	0.9437
	RCAN [52]		25.65M	0.9935	3.4180	0.0157	3.3990	38.0677	0.9509
	SAN [53]		25.58M	0.9934	3.4217	0.0158	3.4251	38.0317	0.9505
	VDSR [29]		8.47M	0.9923	4.2163	0.0175	3.7994	36.7630	0.9436
	DCNN [55]		1.39M	-	4.2000	-	-	35.6600	0.9490
	OISR_LF [40]		24.34M	0.9933	3.4128	0.0159	3.4787	37.9464	0.9508
	OISR_RK2 [40]		24.34M	0.9932	3.4323	0.016	3.4817	37.8845	0.9499
	OISR_RK3 [40]		23.16M	0.9933	3.4388	0.0159	3.4629	37.9137	0.9502
	Dual ODE		12.92M	0.9939	3.2431	0.0155	3.3977	38.0893	0.9527
	Bicubic	8	-	0.9666	5.8962	0.0346	4.2175	30.2056	0.8526
	3DCNN [54]		0.14M	0.9656	9.601	0.0387	8.5924	29.0791	0.8269
	EDSR [30]		26.7M	0.9798	5.2921	0.0286	6.1448	32.4962	0.8795
	GDRRN [14]		0.84M	0.9753	7.4783	0.0321	7.0845	30.9517	0.8582
	RCAN [52]		28.01M	0.9801	5.2484	0.0282	6.0985	32.6003	0.8811
	SAN [53]		27.94M	0.9801	5.1739	0.0282	6.0714	32.6511	0.8818
	VDSR [29]		8.51M	0.9757	7.2839	0.0319	6.8884	31.1479	0.8601
	DCNN [55]		1.54M	-	5.6560	-	-	31.857	0.9020
	OISR_LF [40]		26.7M	0.9798	5.3107	0.0285	6.1195	32.5235	0.8804
	OISR_RK2 [40]		26.7M	0.9798	5.2492	0.0285	6.1272	32.5359	0.8815
	OISR_RK3 [40]		25.52M	0.9796	5.2753	0.0286	6.1647	32.4843	0.8798
	Dual ODE		18.28M	0.9806	4.5781	0.0276	5.9383	32.9072	0.8920

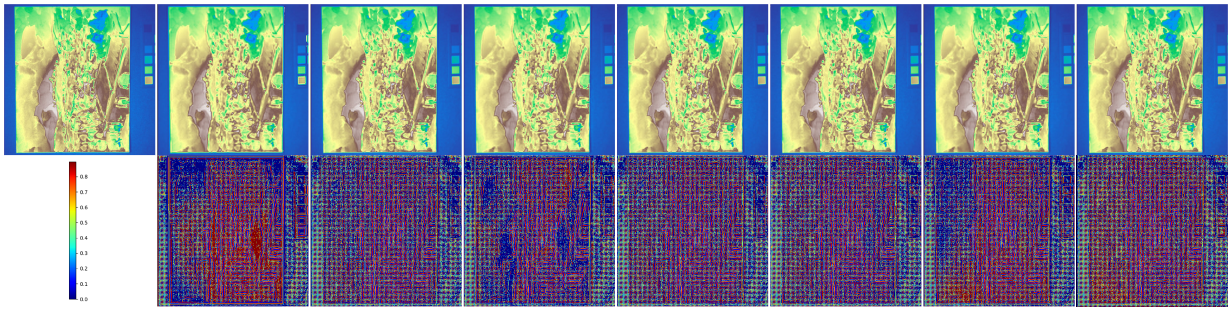


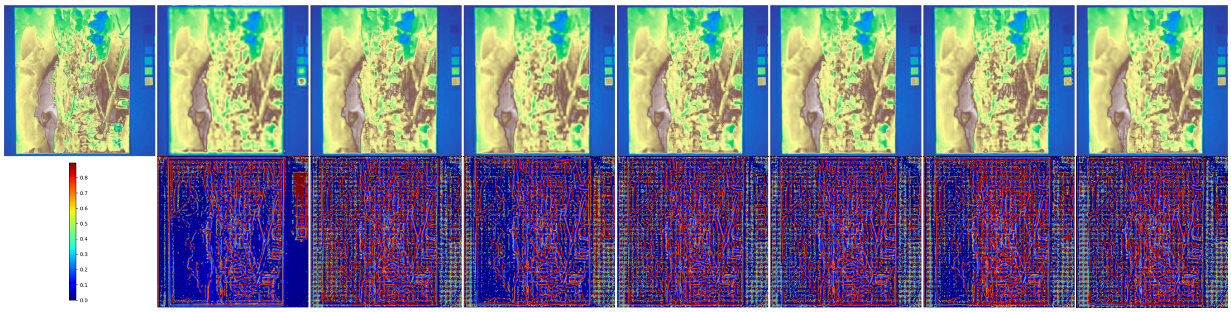
Fig. 2. Qualitative comparisons of our method with other state-of-the-arts on ×4 super-resolution on the CAVE dataset, showing reconstructed images and error maps for each method. Dual ODEs achieve superior performance over other methods in terms of reconstructing highly detailed images with minimal blurring.

performance for each upscaling factor, as shown in Table III. Notably, at the 8 scale, our model achieves a 0.38 dB improvement in PSNR compared to EDSR while using fewer parameters. Furthermore, the advantages of the proposed method become even more apparent when compared to other methods. For instance, compared to the SAN method, our method achieves a PSNR improvement of 0.35 dB with a reduction of 10M parameters. Additionally, the error visualization in the figure clearly illustrates the superior visual effects of the HSI images generated by our Dual ODE method. These images preserve more high-frequency details and exhibit reduced blur compared to the competing methods.

3) Results on the Pavia Centre Dataset: In the Pavia Centre dataset, the results presented in Table IV demonstrate the consistent and robust performance of our Dual ODE model across all upscaling factors. Our method consistently achieves the best reconstruction performance in all cases. Notably, compared to SAN, our method achieves a higher PSNR value by 0.21 (×4) and 0.13 (×8), as well as an improved SSIM by 0.009 (×4) and 0.0117 (×8), all while having nearly half the parameter size. Additionally, the visualization in the figure highlights the effectiveness of our method in reducing reconstruction errors, particularly on the edges of the reconstructed HSI.

4) Results on the Chikusei Dataset: In the Chikusei dataset, the results presented in Table V demonstrate the strong performance of our method in terms of numerical metrics for both 4 and 8 super-resolution scales. Additionally, the visual

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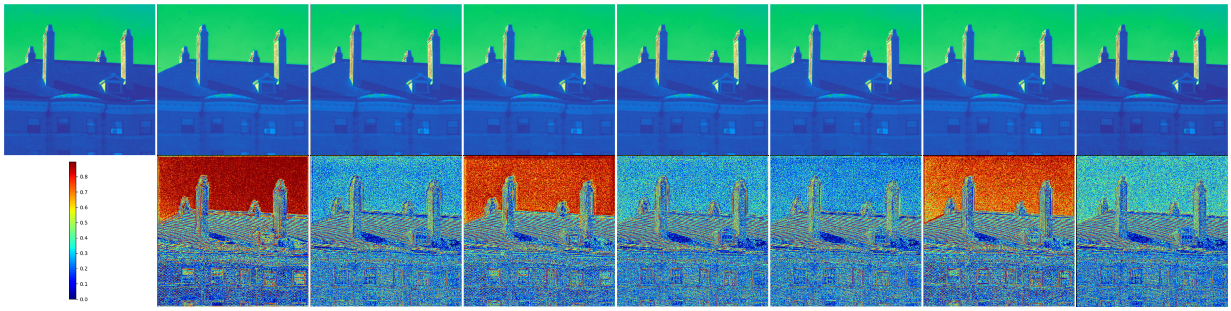
HR	3DCNN [54]	EDSR [30]	GDRNN [14]	RCAN [52]	SAN [53]	VDSR [29]	**Dual ODE**
(PSNR)	(29.0791)	(32.4962)	(30.9517)	(32.6003)	(32.6511)	(31.1479)	(32.9072)
(SSIM)	(0.8269)	(0.8795)	(0.8582)	(0.8811)	(0.8818)	(0.8601)	(0.8920)

Fig. 3. Qualitative comparisons of our method with other state-of-the-arts on ×8 super resolution on the CAVE dataset. The reconstructed images and error maps are displayed for each method. Our Dual ODE approach outperforms the other methods in terms of reconstructing more detailed images with less blurring.

TABLE III

QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH OTHER METHODS ON THE HARVARD DATASET USING CC, SAM, RMSE, ERGAS, PSNR (DB), AND SSIM METRICS. RED INDICATES THE BEST PERFORMANCE

dataset	Model	scale	params	CC ↑	SAM ↓	RMSE ↓	ERGAS ↓	PSNR ↑	SSIM ↑
Harvard	3DCNN [54]	4	0.11M	0.9590	4.3503	0.0183	5.3273	37.4151	0.9051
	EDSR [30]		24.34M	0.9738	3.3235	0.0147	4.1799	39.5998	0.9281
	GDRRN [14]		0.8M	0.9619	4.1219	0.0164	5.0512	38.2700	0.9158
	RCAN [52]		25,65M	0.9742	3.3127	0.0146	4.1445	39.6521	0.9285
	SAN [53]		25.58M	0.9734	3.3500	0.0160	4.2360	39.6113	0.9220
	VDSR [29]		8.47M	0.9674	3.9171	0.0157	4.6534	38.7900	0.9210
	OISR_LF [40]		24.34M	0.9735	3.336	0.0148	4.2118	39.5227	0.9277
	OISR_RK2 [40]		24.34M	0.9742	3.3048	0.0147	4.1554	39.6302	0.9283
	OISR_RK3 [40]		23.16M	0.9743	3.3071	0.0146	4.1388	39.6736	0.9288
	Dual ODE		12.92M	0.9759	3.1628	0.0142	3.9433	39.9828	0.9309
	3DCNN [54]	8	0.14M	0.9242	5.8310	0.0267	7.5562	34.0517	0.8407
	EDSR [30]		26.7M	0.9458	3.8831	0.0224	6.1190	35.9901	0.8646
	GDRRN [14]		0.84M	0.9360	4.7999	0.0236	6.6923	35.2274	0.8558
	RCAN [52]		28.01M	0.9463	3.8697	0.0224	6.0906	36.0125	0.8652
	SAN [53]		27.94M	0.9464	3.8577	0.0224	6.0869	36.0280	0.8654
	VDSR [29]		8.51M	0.9386	4.6859	0.0233	6.5391	35.4154	0.8596
	OISR_LF [40]		26.7M	0.9457	3.9088	0.0225	6.126	35.9872	0.8641
	OISR_RK2 [40]		26.7M	0.9466	3.865	0.0223	6.0729	36.0649	0.8657
	OISR_RK3 [40]		25.52M	0.9468	3.8681	0.0223	6.0788	36.0411	0.8662
	Dual ODE		18.28M	0.9501	3.6235	0.0217	5.8452	36.3797	0.8718



HR	3DCNN [54]	EDSR [30]	GDRNN [14]	RCAN [52]	SAN [53]	VDSR [29]	**Dual ODE**
(PSNR)	(37.4151)	(39.5998)	(38.2700)	(39.6521)	(39.6917)	(38.7900)	(39.9828)
(SSIM)	(0.9051)	(0.9281)	(0.9158)	(0.9285)	(0.9288)	(0.9210)	(0.9309)

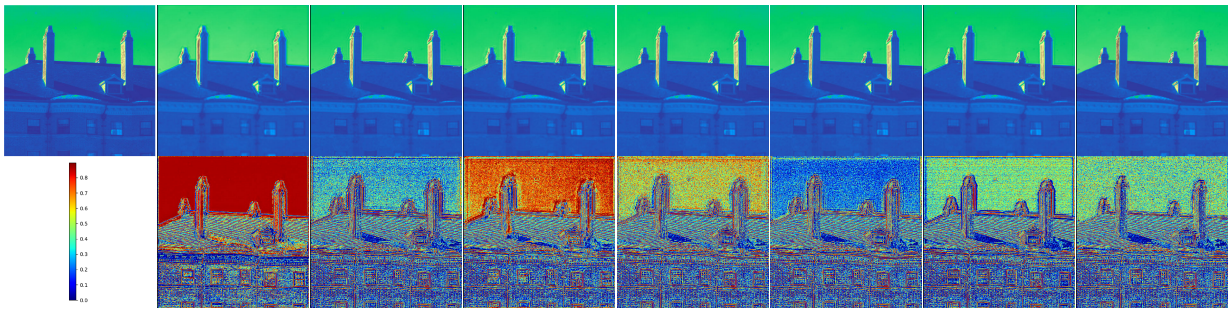
Fig. 4. Qualitative comparisons of our method with other state-of-the-arts on ×4 super-resolution on the Harvard dataset, showing reconstructed images and error maps for each method. Reconstructed images using the Dual ODE approach exhibit superior performance in terms of preserving high-frequency details and reducing blurring compared to other methods.

reconstruction results showcased in Figs. 2–7 further validate the effectiveness of the Dual ODE method in producing high-quality reconstructions.

V. ABLATION STUDY

To gain a deeper understanding and evaluate the effectiveness of our method in HSI super-resolution tasks,

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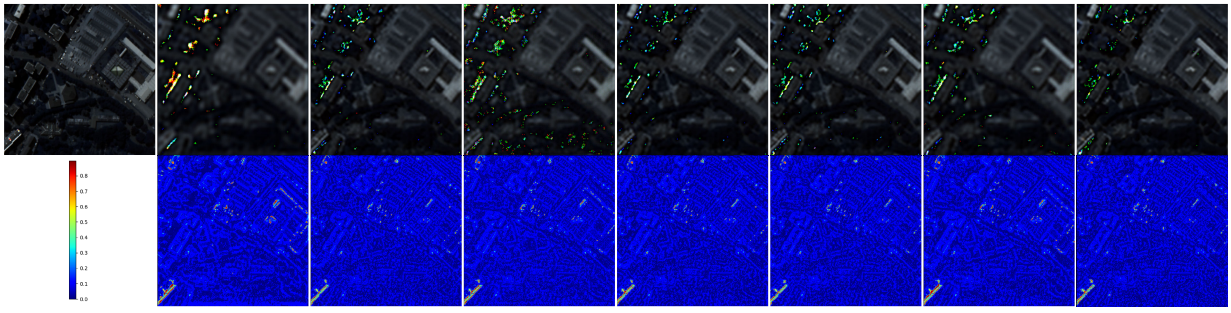
HR	3DCNN [54]	EDSR [30]	GDRNN [14]	RCAN [52]	SAN [53]	VDSR [29]	**Dual ODE**
(PSNR)	(34.0517)	(35.9901)	(35.2274)	(36.0125)	(36.028)	(35.4154)	(36.3797)
(SSIM)	(0.8407)	(0.8646)	(0.8558)	(0.8652)	(0.8654)	(0.8596)	(0.8718)

Fig. 5. Qualitative comparisons of our method with other state-of-the-arts on $\times 8$ super resolution on the Harvard dataset. The reconstructed images and error maps are displayed for each method. The Dual ODE approach produces superior reconstructed images with preserved high-frequency details and reduced blurring compared to other methods.

TABLE IV

QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH OTHER METHODS ON THE PAVIA CENTRE DATASET USING CC, SAM, RMSE, ERGAS, PSNR (DB), AND SSIM METRICS. RED INDICATES THE BEST PERFORMANCE

dataset	Model	scale	params	CC ↑	SAM ↓	RMSE ↓	ERGAS ↓	PSNR ↑	SSIM ↑
Pavia Centre	Bicubic	4	-	0.8594	6.1399	0.0437	6.8814	27.5874	0.6961
	3DCNN [54]		0.79M	0.8256	9.8870	0.0534	9.6806	25.8129	0.6295
	EDSR [30]		24.66M	0.8935	7.9330	0.0418	7.7050	27.8607	0.7667
	GDRRN [14]		2.40M	0.8678	10.1347	0.0464	8.5384	26.9696	0.7128
	RCAN [52]		15.88M	0.8911	8.0033	0.0423	7.7906	27.7604	0.7612
	SAN [53]		15.87M	0.8948	8.0091	0.0416	7.6613	27.9088	0.7685
	VDSR [29]		9.48M	0.8621	8.9933	0.0478	8.6892	26.7684	0.6952
	OISR_LF [40]		24.66M	0.8953	8.0614	0.0414	7.6426	27.9538	0.7739
	OISR_RK2 [40]		24.66M	0.8941	8.0074	0.0417	7.6841	27.8847	0.7686
	OISR_RK3 [40]		23.48M	0.8945	7.9192	0.0417	7.672	27.8971	0.7689
	Dual ODE		13.18M	0.9003	7.3908	0.0407	7.4601	28.1193	0.7775
	Bicubic	8	-	0.6969	7.8478	0.0630	4.8280	24.5972	0.4725
	3DCNN [54]		1.16M	0.7208	11.5017	0.0662	11.9214	23.9815	0.4781
	EDSR [30]		17.58M	0.7801	9.8351	0.0597	10.6890	24.9236	0.5547
	GDRRN [14]		2.77M	0.7596	10.9731	0.0619	11.1338	24.5755	0.5194
	RCAN [52]		18.24M	0.7756	9.7988	0.0603	10.7794	24.8443	0.5453
	SAN [53]		18.23M	0.7768	9.7574	0.0601	10.7571	24.8646	0.5483
	VDSR [29]		9.86M	0.7666	10.2380	0.0613	10.9734	24.6920	0.5305
	OISR_LF [40]		27.02M	0.7784	9.9482	0.0598	10.7291	24.8963	0.555
	OISR_RK2 [40]		27.02M	0.7776	9.7642	0.06	10.7335	24.8819	0.5498
	OISR_RK3 [40]		25.84M	0.7783	9.7728	0.06	10.7263	24.8893	0.5505
	Dual ODE		15.54M	0.7833	9.0163	0.0594	10.6291	24.9695	0.5600

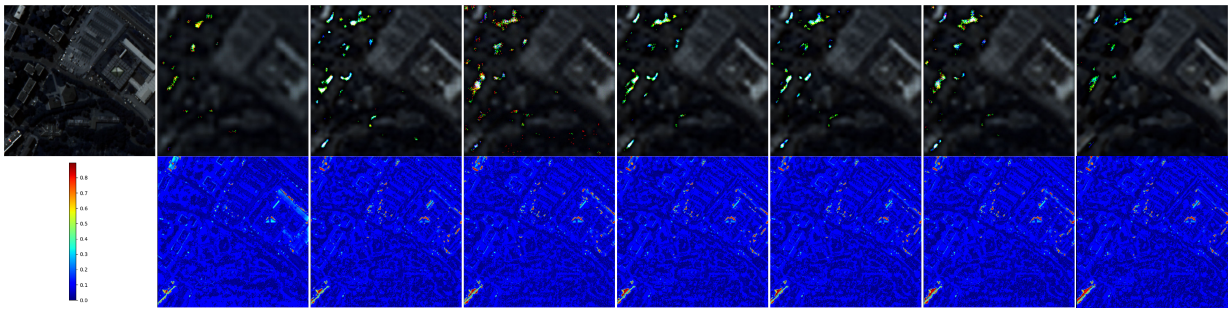


HR	3DCNN [54]	EDSR [30]	GDRNN [14]	RCAN [52]	SAN [53]	VDSR [29]	**Dual ODE**
(PSNR)	(25.8129)	(27.8607)	(26.9696)	(27.7604)	(27.9088)	(26.7684)	(28.1193)
(SSIM)	(0.6295)	(0.7667)	(0.7128)	(0.7612)	(0.7685)	(0.6952)	(0.7775)

Fig. 6. Qualitative comparisons of our method with other state-of-the-arts on $\times 4$ super-resolution on the Pavia Centre dataset, showing reconstructed images and error maps for each method. Comparison of our Dual ODE approach with other state-of-the-art methods demonstrates its superior ability to reconstruct images with higher level of details and reduced blurring.

we conducted a series of ablation experiments. The Harvard dataset contains numbers of HSI data, which is more suitable for assessing model performance and thus has been widely utilized [58]. For this purpose, we choose the Harvard dataset for the ablation study. The primary objective of these experiments was to scrutinize the influence of various components

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HR	3DCNN [54]	EDSR [30]	GDRNN [14]	RCAN [52]	SAN [53]	VDSR [29]	**Dual ODE**
(PSNR)	(23.9815)	(24.9236)	(24.5755)	(24.8443)	(24.8646)	(24.6920)	(24.9695)
(SSIM)	(0.4781)	(0.5547)	(0.5194)	(0.5453)	(0.5483)	(0.5305)	(0.5600)

Fig. 7. Qualitative comparisons of our method with other state-of-the-arts on $\times 8$ super resolution on the Pavia dataset. The reconstructed images and error maps are displayed for each method. Comparison of reconstructed images using the Dual ODE approach with other methods demonstrates its superiority in preserving high-frequency details and reducing blurring.

TABLE V

QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH OTHER METHODS ON THE CHIKUSEI DATASET USING CC, SAM, RMSE, ERGAS, PSNR (DB), AND SSIM METRICS. RED INDICATES THE BEST PERFORMANCE

dataset	Model	scale	params	CC ↑	SAM ↓	RMSE ↓	ERGAS ↓	PSNR ↑	SSIM ↑
Chikusei	Bicubic	4	-	0.9212	3.4040	0.0156	6.7564	37.6377	0.8949
	3DCNN [54]		1.22M	0.9301	3.3402	0.0145	6.3809	38.2271	0.9103
	EDSR [30]		15.34M	0.9454	2.6215	0.0128	5.5468	39.3388	0.9311
	GDRRN [14]		2.36M	0.9501	2.6361	0.0121	5.3773	39.7589	0.9373
	RCAN [52]		26.1M	0.9492	2.6043	0.0123	5.3706	39.6861	0.9359
	SAN [53]		26.03M	0.9463	2.5829	0.0127	5.5026	39.4203	0.9324
	VDSR [29]		5.31M	0.9328	2.9526	0.0139	6.3163	38.4218	0.9183
	OISR_LF [40]		24.78M	0.9493	2.6193	0.0122	5.4006	39.6828	0.936
	OISR_RK2 [40]		24.78M	0.9462	2.5928	0.0127	5.5058	39.4154	0.9321
	OISR_RK3 [40]		23.6M	0.9463	2.5901	0.0127	5.5055	39.4136	0.9323
	Dual ODE		13.59M	0.9568	2.3860	0.0114	4.9378	40.3679	0.9451
	Bicubic	8	-	0.8314	5.0436	0.0224	4.8488	34.5049	0.8288
	3DCNN [54]		1.81M	0.8320	5.3230	0.0219	9.8466	34.5214	0.8118
	EDSR [30]		17.7M	0.8521	4.6776	0.0210	8.9770	35.0921	0.8299
	GDRRN [14]		2.95M	0.8598	4.5838	0.0201	8.9571	35.2959	0.8378
	RCAN [52]		28.46M	0.8589	4.6760	0.0201	8.9825	35.2770	0.8387
	SAN [53]		28.39M	0.8547	4.5732	0.0208	8.9065	35.1628	0.8319
	VDSR [29]		5.9M	0.8466	4.7064	0.0210	9.3637	34.9213	0.8301
	OISR_LF [40]		27.14M	0.8507	4.761	0.021	9.0188	35.0542	0.8281
	OISR_RK2 [40]		27.14M	0.8569	4.601	0.0207	8.8547	35.2278	0.8343
	OISR_RK3 [40]		25.96M	0.8568	4.5632	0.0207	8.8477	35.2209	0.8341
	Dual ODE		15.95M	0.8781	3.9402	0.0191	8.2487	35.8694	0.8567

on the model's performance and identify the optimal parameter configurations.

A. System Dynamics f Number

We conducted an ablation study on the Harvard dataset at the 4 super-resolution scale. The results are presented in Table VI. As the number of f [equivalent to k in (3)] increases, the parameter size of the model also increases, and the overall reconstruction performance improves gradually. However, to strike a balance between reconstruction performance and model size, we selected the optimal number of f.

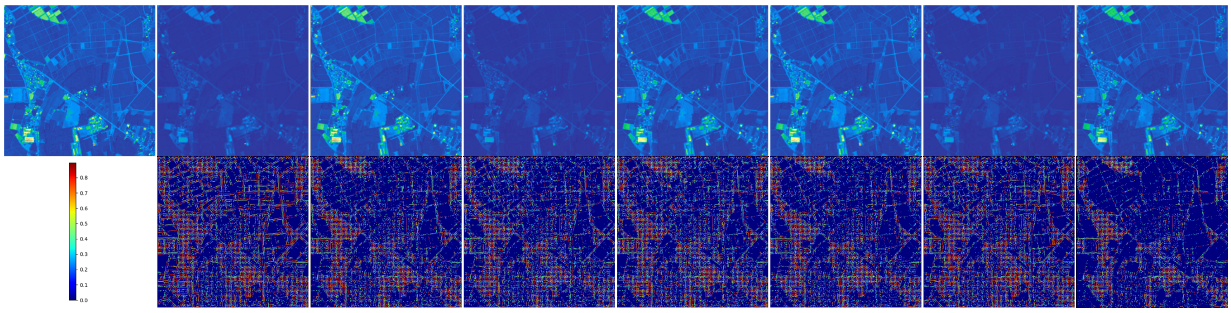
B. Time-Step Effects

We performed an ablation study on the Harvard datasets at the 8 super-resolution scale, and the results are presented in Table VII. It can be observed that as the time step of the ODE solver increases, the computational cost of the model decreases. However, it is important to note that the optimal time step varies for different datasets and super-resolution scales. For $\times 8$ scale super-resolution on the Harvard dataset, the best performance is achieved with a time step of 0.05. These findings highlight the significance of selecting an appropriate time step that balances computational efficiency and super-resolution performance, taking into consideration the specific dataset and super-resolution scale.

C. Higher-Order Method Effects

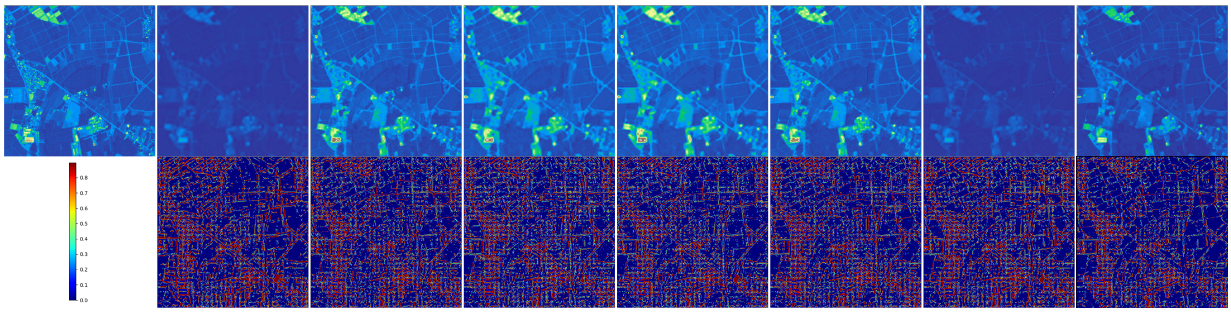
We conducted an ablation study on the Harvard datasets at the 8 super-resolution scale. The results are presented in Table VIII. It can be observed that as the order of the Dual ODE solver increases, the number of model parameters also increases. However, the super-resolution performance does not consistently improve with the increase in solver order. For the 8 super-resolution scale on the Harvard dataset, using the Euler method as the solver yields the optimal results. These

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HR	3DCNN [54]	EDSR [30]	GDRNN [14]	RCAN [52]	SAN [53]	VDSR [29]	**Dual ODE**
(PSNR)	(38.2271)	(39.3388)	(39.7589)	(39.6861)	(39.4203)	(38.4218)	(40.3679)
(SSIM)	(0.9103)	(0.9311)	(0.9373)	(0.9359)	(0.9324)	(0.9183)	(0.9451)

Fig. 8. Qualitative comparisons of our method with other state-of-the-arts on $\times 4$ super-resolution on the Chikusei dataset, showing reconstructed images and error maps for each method. Our Dual ODE approach produces more detailed images with less blurring, outperforming other methods in terms of image reconstruction quality.



HR	3DCNN [54]	EDSR [30]	GDRNN [14]	RCAN [52]	SAN [53]	VDSR [29]	**Dual ODE**
(PSNR)	(34.5214)	(35.0921)	(35.2959)	(35.277)	(35.1628)	(34.9213)	(35.8694)
(SSIM)	(0.8118)	(0.8299)	(0.8378)	(0.8387)	(0.8319)	(0.8301)	(0.8567)

Fig. 9. Qualitative comparisons of our method with other state-of-the-arts on $\times 8$ super resolution on the Chikusei dataset. The reconstructed images and error maps are displayed for each method. Our Dual ODE approach results in reconstructed images with superior performance in terms of preserving high-frequency details and reducing blurring compared to other methods.

TABLE VI

QUANTITATIVE COMPARISON OF METHODS WITH DIFFERENT f NUMBERS FOR THE 4 UPSAMPLING FACTOR OVER THE HARVARD DATASET

dataset	f number (k)	params	CC ↑	SAM ↓	RMSE ↓	ERGAS ↓	PSNR ↑	SSIM ↑
Harvard	1	7.63M	0.9756	3.1717	0.0143	3.9675	39.9179	0.9301
	2	10.27M	0.9759	3.1632	0.0142	3.9441	39.9731	0.9305
	3	12.92M	0.9759	3.1628	0.0142	3.9433	39.9828	0.9309
	4	15.57M	0.9760	3.1588	0.0142	3.9498	39.9791	0.9307
	5	18.22M	0.9759	3.1573	0.0142	3.9453	39.9830	0.9306
	6	20.87M	0.9759	3.1566	0.0142	3.9411	39.9958	0.9309

TABLE VII

QUANTITATIVE COMPARISON OF DIFFERENT ODE SOLVER TIME-STEP FOR THE 8 UPSAMPLING FACTOR OVER THE HARVARD DATASET

dataset	params	time step	flops (G)	CC ↑	SAM ↓	RMSE ↓	ERGAS ↓	PSNR ↑	SSIM ↑
Harvard	15.28M	0.05	14421.40	0.9505	3.6147	0.0216	5.8182	36.4135	0.8721
		0.1	7655.46	0.9501	3.6235	0.0217	5.8452	36.3797	0.8718
		0.125	6302.27	0.9503	3.6182	0.0216	5.8207	36.4129	0.8721
		0.2	4272.48	0.9502	3.6208	0.0216	5.8317	36.3989	0.8719
		0.5	2242.70	0.9498	3.6369	0.0218	5.8596	36.3461	0.8711
		1.0	1566.11	0.9495	3.6433	0.0219	5.8815	36.2918	0.8702

experimental findings validate the discussion in Section III-D, emphasizing the importance of carefully selecting the appropriate ODE solver with the proper order based on the specific problem and context.

D. With or Without ODE Effects

We conducted an ablation study on the Harvard dataset, evaluating the performance of our method at two different super-resolution scales. The numerical results are summarized

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TABLE VIII
QUANTITATIVE COMPARISON OF DIFFERENT DUAL ODE NUMERICAL SOLVER FOR THE 8 UPSAMPLING FACTOR OVER THE HARVARD DATASET

dataset	method	params	CC ↑	SAM ↓	RMSE ↓	ERGAS ↓	PSNR ↑	SSIM ↑
Harvard	Dual ODE_Euler	15.28M	0.9806	4.5781	0.0276	5.9383	32.9072	0.8920
	Dual ODE_RK2	23.23M	0.9810	4.9030	0.0272	5.9443	32.8881	0.8942
	Dual ODE_RK3	31.08M	0.9789	4.5031	0.0288	6.1372	32.6501	0.8859

TABLE IX
QUANTITATIVE COMPARISON OF WITH/WITHOUT ODE METHODS FOR THE 4 UPSAMPLING FACTOR OVER THE HARVARD DATASET

dataset	method	params	CC ↑	SAM ↓	RMSE ↓	ERGAS ↓	PSNR ↑	SSIM ↑
Harvard	ODE-spectral	12.91M	0.9759	3.1630	0.0142	3.9505	39.9725	0.9307
	spatial-ODE	12.91M	0.9756	3.1724	0.0144	3.9871	39.8853	0.9294
	Dual ODE	12.92M	0.9759	3.1628	0.0142	3.9433	39.9828	0.9309

TABLE X
QUANTITATIVE COMPARISON OF SPATIAL–SPECTRAL RESIDUAL NETWORKS AND SPATIAL–SPECTRAL DUAL ODE FOR THE 8 UPSAMPLING FACTOR OVER THE HARVARD DATASET

dataset	Model	params	CC ↑	SAM ↓	RMSE ↓	ERGAS ↓	PSNR ↑	SSIM ↑
Harvard	SSPSR [51]	28.46M	0.9499	3.6146	0.0218	5.8586	36.3510	0.8715
	Dual ODE	18.28M	0.9501	3.6235	0.0217	5.8452	36.3797	0.8718

in Table IX. The findings indicate that both spatial and spectral ODEs contribute to improving the super-resolution performance. Notably, the inclusion of the spectral ODE leads to a more significant improvement compared to the spatial ODE alone. Moreover, the combination of both spatial and spectral ODEs, known as the dual ODE, achieves the optimal super-resolution effect. These results highlight the importance of incorporating both spatial and spectral dynamics in the HSI super-resolution task.

E. Replacing Residual Networks With the Dual ODE

Our final ablation study focuses on the performance comparison between the traditional handcrafted spatial–spectral residual networks and the ODE-based spatial–spectral network designs. As a baseline, we selected the most relevant and state-of-the-art study (SSPSR) [51], which employs spatial–spectral residual structures. As shown in Table X, our Dual ODE method consistently outperforms the spatial–spectral residual networks with fewer parameters in terms of PSNR, SSIM, and RMSE.

These results demonstrate the superior performance of our Dual ODE method over the traditional spatial–spectral residual networks. Our approach achieves higher reconstruction quality while effectively reducing model complexity. These findings validate the effectiveness and efficiency of leveraging ODE-based spatial–spectral network designs in HSI super-resolution tasks.

VI. CONCLUSION

In this article, we have presented the Dual ODE approach as a new method for HSI super-resolution. Our approach harnesses the power of ODEs to design a spectral–spatial feature extraction network that effectively captures both spatial and spectral information. The key innovation of the Dual ODE lies in the incorporation of a spatial ODE block for spatial feature extraction and a spectral ODE block for internal spectral feature representation. By redefining the traditional residual module using multiple ODE functions, we enhance the network's ability to extract informative features. Through extensive experimentation, we have demonstrated the superior performance of our Dual ODE approach in HSI super-resolution. Our method outperforms state-of-the-art models, striking a balance between reconstruction performance and model complexity. It provides a principled and reliable framework for designing HSI super-resolution networks, offering enhanced feature representations that lead to improved results.

The insights gained from our study shed light on the underlying mechanism of residual networks and contribute to the advancement of HSI super-resolution techniques. Our approach not only provides efficient mechanisms for designing HSI super-resolution networks, but also offers valuable guidelines for future research in this field. By leveraging the dynamics of ODEs, we open up new avenues for exploring the potential of HSI super-resolution and encourage further advancements in this exciting area of research.

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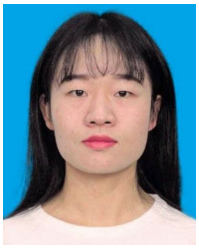
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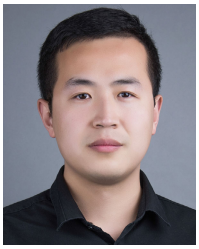


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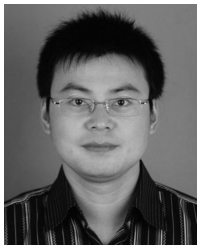


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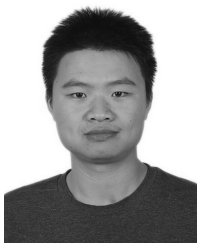
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